

2025 CNS Lab – Paper Review Seminar

Ignore Me But Don't Replace Me

: Utilizing Non-Linguistic Elements for Pretraining on the Cybersecurity Domain

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Speaker : **Seungho Song**

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Disclaimer : This slide was based on a document from the original author. (Eugene Jang)

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Introduction

NLP for Cybersecurity

- Cybersecurity defense requires Cyber Threat Intelligence (CTI)
 - **Actionable information** of cyber threats
 - Requires manual, expert labor
- NLP methods can help deal with:
 - Constantly evolving information
 - Make timely decisions



Introduction

NLP for Cybersecurity - Challenges

- Cybersecurity documents:
 - Require domain expertise
 - Contain large amounts of non-linguistic elements (Hash values, URLs, etc)

The main functionality of cblRWuoCc.so

The file **miner64_s** (8d8bdb58c5e57c565542040ed1988af9 — RiskTool.Linux.BitCoinMiner.a) downloaded in such a way is stored in **/tmp/m** on the victim's system.

Also, we discovered that the DTrack malware (MD5 739812e2ae1327a94e441719b885bd19) employs the same shellcode loader as "Backdoor.Preft" malware (MD5 2f553cba839ca4dab201d3f8154bae2a), [published/reported by Symantec](#) – note that Symantec recently described the Backdoor.Preft malware as "aka Dtrack, Valefor". Apart from the code similarity, the actor used 3Proxy tool (MD5 5bc4b606f4c0f8cd2e6787ae049bf5bb), and that tool was also previously employed by the Andariel/StoneFly/Silent Chollima group (MD5 95247511a611ba3d8581c7c6b8b1a38a). Symantec attributes StoneFly as the North Korean-linked actor behind the DarkSeoul incident.

known for been used to infect **IOT devices** and **OSX systems** in the past. The source is **publicly available** and could be used by anyone. This sample (d8e93252f41e8b0d10cffa92923eeab94c6c42e8acc308e91340df102042c8c8) is configured with hardcoded c2 irc server (asdgds.uselesslongdomain[.]info). We were able to connect to the live irc server using credentials hard coded in the



Introduction

NLP for Cybersecurity - Challenges

- Cybersecurity texts:
 - **Require domain expertise**
-> **can be addressed** with domain pretraining
 - **Contain large amounts of non-linguistic elements (Hash values, URLs, etc)**
-> **interferes** with domain pretraining

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Introduction

Pretraining with Cybersecurity Text

- MLM teaches models to reconstruct a **masked token** into the **original token**.
- But some tokens have **no linguistic meaning**, predicting these are **not effective**.
- We could just **replace** them but...
 - Loses information
 - Model doesn't learn to utilize or recognize NLEs

This sample (d8e93252f41e8b0d10cffa92923
eeab94c6c42e8acc308e91340df102042c8c8 SHA)
is configured with hardcoded c2 irc server
(asdgsd.uselesslongdomain[.]info URL).

This	sample	(	d	8	e	93	252	f	41		
e	8	b	0	d	10	c	ff	a	9	29	23
ee	ab	94	c	6	c	42	e	8	acc	308	
e	9	13	40	df	10	10	20	42	c	8	
c	e	)	is	configured	with	hard	coded				
c2	irc	server	(	as	d	gs	d	.	u		
seless	long	domain	[	.	]	info	)	.			

Method

Identifying non-linguistic elements

We manually identify **common archetypes** of non-linguistic elements (NLEs) found in cybersecurity texts.

NLE	Ours (PS)	Ours (Full)	Wiki	C4
URL	16,272	5,172	62	404
EMAIL	3,282	901	< 1	33
IP	2,503	780	3	15
MD5	2,651	754	< 1	1
SHA	550	161	< 1	< 1
BTC	1,024	273	< 1	< 1
CVE	1,225	550	< 1	3

- Our cybersecurity corpus has much more NLEs compared to other pretraining corpora.
- Pretraining Subset: To better study their influence, we experiment with a subset of sources with more NLEs.

Method

Distinction between SLEs and FNLEs

- We make one extra distinction between two groups.
- Semi-linguistic Elements:
 - **URLs, emails**
 - Sometimes informative
 - github[.]url-mini[.]com/msg.zip
 - Sometimes random
 - asdgsd.uselesslongdomain[.]info
- Fully Non-linguistic Elements:
 - **IP, MD5, SHA, BTC, CVE**
 - Always random (generated by protocols)

SLE

FNLE

NLE	Ours (PS)	Ours (Full)	Wiki	C4
URL	16,272	5,172	62	404
EMAIL	3,282	901	< 1	33
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Method

Modifying Pretraining – Vanilla MLM

- 15% of tokens are randomly selected
- The selected tokens are masked (gray bars)
- The model attempts to guess the original tokens for each masked token

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c	←	)	is	configured	with	←	hard	←	coded				
c2	irc	server	(	as	←	gs	d	.	u				
seless	long	domain	←	.	]	info	)	.					

Method

Modifying Pretraining – Marking tokens

- Label original text with NLE spans
- We can use this information to guide training

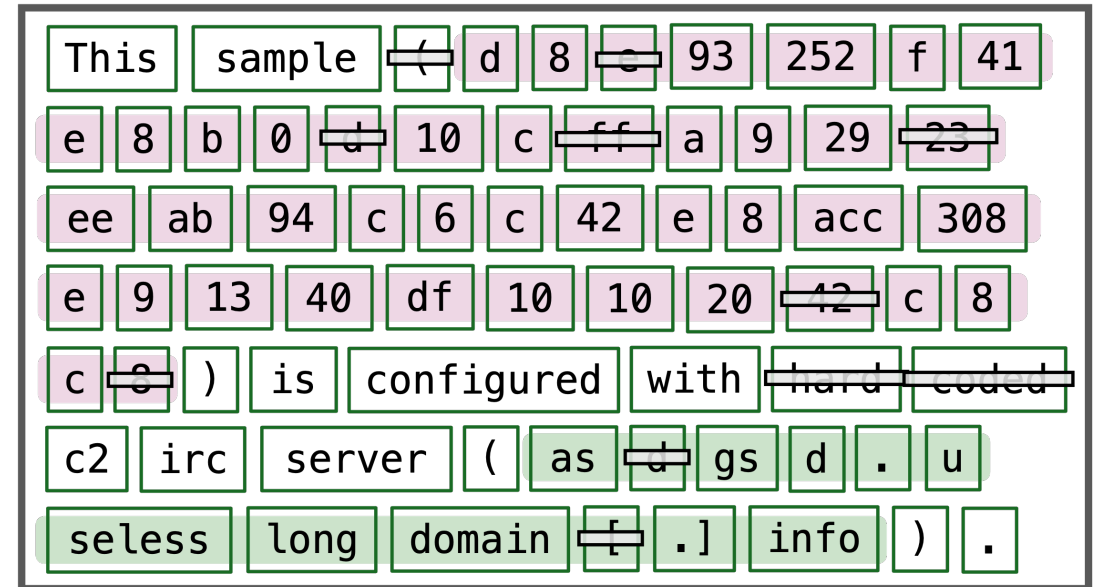
This sample (d8e93252f41e8b0d10cffa92923
eeab94c6c42e8acc308e91340df102042c8c8 SHA)
is configured with hardcoded c2 irc server
(asdgsd.uselesslongdomain[.]info URL).

This sample (d 8 e 9 3 2 5 2 f 4 1
e 8 b 0 d 1 0 c f f a 9 2 9 2 3
ee ab 9 4 c 6 c 4 2 e 8 acc 3 0 8
e 9 1 3 4 0 d f 1 0 1 0 2 0 4 2 c 8
c) is configured with hardcoded
c2 irc server (as d g s d . u
seless long domain .] info) .

Method

Modifying Pretraining – 1) Selective Masking

- We can choose NOT to mask if it is part of a NLE span.
- We can do this for...



Method

Modifying Pretraining – 1) Selective Masking

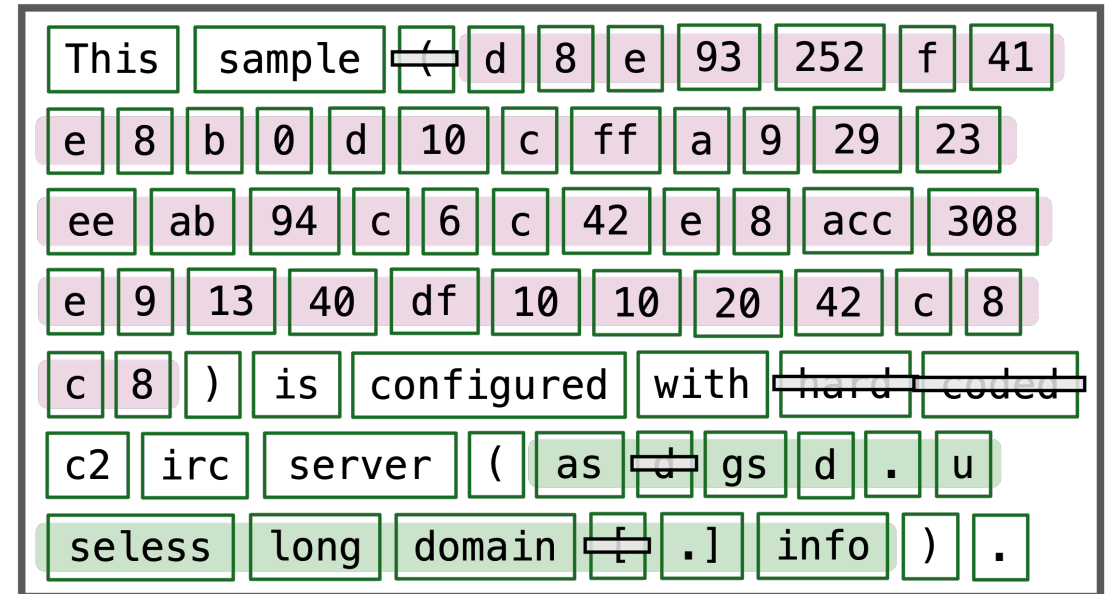
- We can choose NOT to mask if it is part of a NLE span.
- We can do this for...
 - all NLEs

This sample ← d 8 e 93 252 f 41
e 8 b 0 d 10 c ff a 9 29 23
ee ab 94 c 6 c 42 e 8 acc 308
e 9 13 40 df 10 10 20 42 c 8
c 8) is configured with ~~hard-coded~~
c2 irc server (as d gs d . u
seless long domain [.] info) .

Method

Modifying Pretraining – 1) Selective Masking

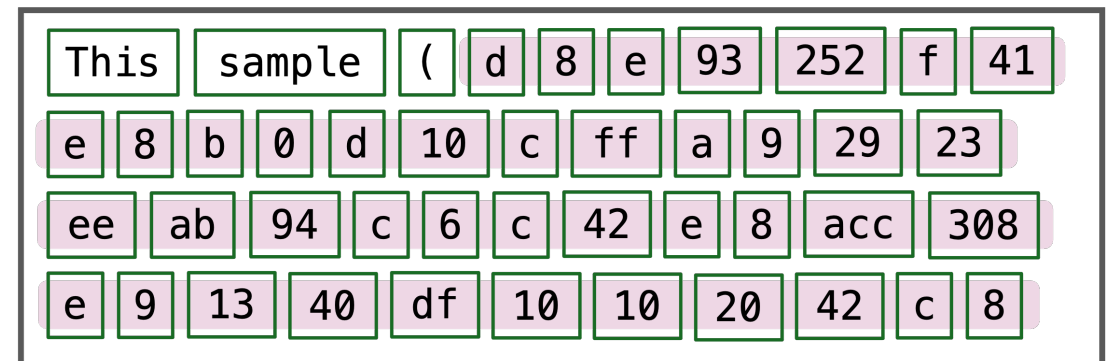
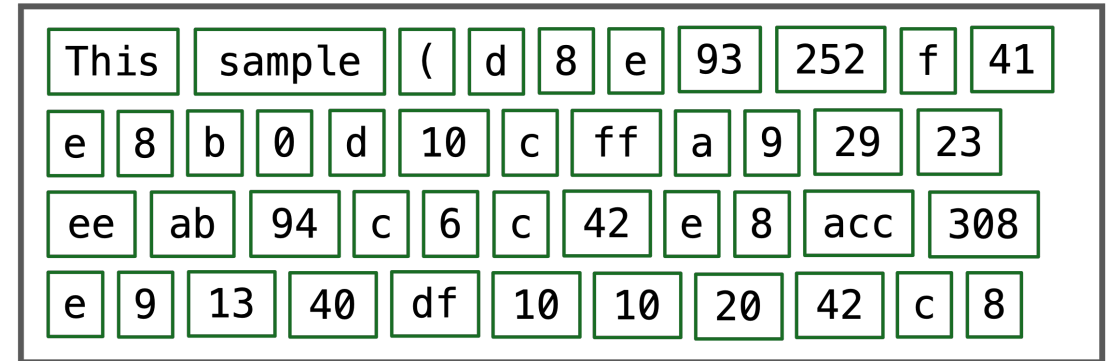
- We can choose NOT to mask if it is part of a NLE span.
- We can do this for...
 - all NLEs
 - Or just the random ones (FNLEs).



Method

Modifying Pretraining – 2) NLE Classification

- We can make the model predict for NLE tokens.
- 8-class classification problem
 - (7 NLE types)
- Auxiliary task alongside MLM task.



Pretraining Experiments

Pretraining Strategies

Strategies	Masks		NLEC	Example (MLM)	Example (NLE Classification)
	SLE	FNLE			
Vanilla MLM	✓	✓		The Dropper drops a zipped SysJoker (53f1bb23f670d331c9041748e7e8e396) from C2 https[://]github[.]url-mini[.]com/msg.zip, copies it to	N/A
Replace All		(replaced)		The Dropper drops a zipped SysJoker (<MD5>) from C2 <URL>, copies it to	N/A
Vanilla + NLEC	✓	✓	✓	The Dropper drops a zipped SysJoker (53f1bb23f670d331c9041748e7e8e396) from C2 https[://]github[.]url-mini[.]com/msg.zip, copies it to	The Dropper drops a zipped SysJoker (53f1bb23f670d331c9041748e7e8e396) from C2 https[://]github[.]url-mini[.]com/msg.zip, copies it to
Mask-Semis	✓			The Dropper drops a zipped SysJoker (53f1bb23f670d331c9041748e7e8e396) from C2 https[://]github[.]url-mini[.]com/msg.zip, copies it to	N/A
Mask-Semis + NLEC	✓		✓	The Dropper drops a zipped SysJoker (53f1bb23f670d331c9041748e7e8e396) from C2 https[://]github[.]url-mini[.]com/msg.zip, copies it to	The Dropper drops a zipped SysJoker (53f1bb23f670d331c9041748e7e8e396) from C2 https[://]github[.]url-mini[.]com/msg.zip, copies it to
Mask-None + NLEC			✓	The Dropper drops a zipped SysJoker (53f1bb23f670d331c9041748e7e8e396) from C2 https[://]github[.]url-mini[.]com/msg.zip, copies it to	The Dropper drops a zipped SysJoker (53f1bb23f670d331c9041748e7e8e396) from C2 https[://]github[.]url-mini[.]com/msg.zip, copies it to

Pretraining Experiments

Experiments

- Downstream Tasks : To Evaluate the **applicability of PLMs**
 - Fine-tuned using below datasets
 - CyNER : NER dataset of annotated malware threat reports.
 - CySecED : Event detection dataset of annotated articles from The Hacker News.
 - MalwareTextDB (MTDB) : Dataset of malware reports annotated for 4 types in MalwareTextDB
- Probing Tasks : To evaluate the **performance of model weights**
 - Identify 226 tokens from MITRE's database.
 - 77,983 tokens were masked in Validation corpus.
 - 4,906 tokens were **near FNLEs**. (also mark for NLE evaluation)

Pretraining Experiments

Experiments

Strategies	Downstream Tasks			Probing Tasks	
	CyNER	CySecED	MTDB	All	Near-FNLEs
RoBERTa	0.637	0.504	0.802	0.278	0.270
Vanilla MLM	0.648	0.510	0.822 [†]	0.382	0.460
Replace All	<u>0.664[†]*</u>	<u>0.544</u>	<u>0.827[†]</u>	0.381	0.438
Vanilla + NLEC	0.652	0.526	0.820	0.380	0.455
Mask-Semis	0.638	0.538	0.817	0.386	<u>0.463</u>
Mask-Semis + NLEC	0.667[†]*	0.544[†]	0.825 [†]	<u>0.383</u>	0.464
Mask-None + NLEC	0.643	0.533 [†]	0.829[†]	0.382	0.452

- NLEC helps, but **not significantly on its own**
- Selective MLM helps, especially with NLEC. (Mask-Semis + NLEC)
- Replacement is good for downstream tasks, but hurts probing performance.

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- NLEC helps, but **not significantly on its own**
- Selective MLM helps, especially with NLEC. (Mask-Semis + NLEC)
- Replacement is good for downstream tasks, but **hurts probing performance**.
 - Especially worst in Near-FNLEs.

CyBERTuned Experiments

CyBERTuned

- Using the best-performers strategy: *Mask-Semis + NLEC*
- Downstream Tasks (on a wider variety of cybersecurity tasks)
 - CASIE : Event detection dataset of annotated [news articles](#)
 - TwitterThreats (TT) : Biseq classification dataset of annotated tweets that [mentioned threat keywords](#)
 - CYDEC : Biseq classification dataset of annotated tweets that [mentioned cybersecurity keywords](#)
- Baseline
 - RoBERTa-base (2019)
 - CyBERT (2021)
 - CySecBERT (2022)
 - SecureBERT (2022)



CyBERTuned Experiment

CyBERTuned Evaluations

	Token Class.			Sequence Class.		MCQA
	CASIE	CyNER	CySecED	CYDEC	TT	MTDB
RoBERTa-base	0.748	0.637	0.504	<u>0.829</u>	0.831	0.802
CyBERT	0.711 [†]	0.462 [†]	0.361 [†]	0.798	0.832	0.731 [†]
CySecBERT	0.734	0.572 [†]	0.491	0.814	<u>0.845[†]</u>	0.808
SecureBERT	0.753	<u>0.638</u>	<u>0.529[†]</u>	0.816	0.828	<u>0.825</u>
CyBERTuned (Ours)	<u>0.750</u>	0.654	0.585[†]	0.844	0.857[†]	0.861[†]

- CyBERTuned performs consistently well across tasks.
- On Token-level tasks, BERT-based models struggle, **even losing to RoBERTa-base.**
 - Possibly due to BERT's aggressive pre-tokenization.

Discussion

RoBERTa's performance

Usually achieved good performance, often **outperforming domain-pretrained models**.

-> additional **domain-specific knowledge doesn't necessarily improve** performance in Cybersecurity domain.

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Human scores

- CYDEC : 0.59

- TwitterThreats : 0.66

-> **Already exceeded human performances.**

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Discussion

NLE Classification as auxiliary task

NLE classification loss plateaus quite early during pre-training.

Suggested solutions : **Drop NLE Classification task** after loss plateaus. (will investigated in further study)

NLEs of other domains

NLE in Cybersecurity domain : containing informative content (SLEs)

-> performance of NLE classification **depends on informational contents of NLEs.**

Limitation

NLE Types

Limited to types that are easily identifiable. -> Future : will detect more complex NLEs (code, filepath, etc).

Downstream tasks

Many factors to consider when fine-tuning downstream tasks
-> inconsistent runs

NLEs in different domains

Investigated only for cybersecurity domain.



Conclusion

- Develop a method to pretrain models on the cybersecurity domain.
- Compared to replacing, this method
 - Performs better on downstream tasks
 - Preserves NLEs in training
- CyBERTuned is more capable of utilizing information in NLEs

The PDF contains an action object. Upon a victim opening the PDF it will send a query to Google: `http://www[.]google[.]com/url?q=http%3A%2F%2F934824324938247%2F2U9sSOJYzfUQqw`. This link is a typical <mask> attack.

Compute

Computation time on cpu: 0.164 s

redirect

0.170

